

Q and P: corrected implementation of a CVaR surrogate

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The pair in two sentences

Q, *A Systematic Approach to Mechanism Design with Stochastic Dynamic Stability*, proposes quadratic payments for strategic resource allocation, claiming welfare implementation, budget balance, individual rationality, and mean-square convergence of variable sample-size proximal best responses. P, *Distributed risk-averse optimization via CVaR*, supplies empirical conditional value-at-risk (CVaR) values and a one-point gradient estimator for cooperative optimization, whose fixed-batch limit optimizes an expected empirical smoothed surrogate (ledger entries 3, 13, 15, 19).

The connexion pursued

The initial connexion was to use P's value oracle in Q's strategic learning scheme. The supported replacement is narrower: correct Q's payments, implement a fixed risk surrogate, and quantify the difference from population-risk incentives. A weighted proximal-point learner replaces Q's original iteration (entries 15, 17, 19, 24, 26). This article consolidates the ledger through entry 27 and both peers' notes; the allowance ended without a final common readiness checkpoint covering every last entry. Earlier readiness declarations are not treated as approval of later material.

What was found and why

An obstruction preceding CVaR. Emmy and Ada independently found that Q's implementation equality P2(i) contradicts its strict LMI P1(ii) (entries 1–3). For a nonzero resource vector v , put $h = \text{col}((0, v), \dots, (0, v))$. Q's price blocks give

$$h^\top (\Psi + \Psi^\top) h = -2 \sum_n v^\top \left(\sum_m A_{nm}^n \right) v = 0,$$

whereas the strict certificate requires a value at most $-vN\|v\|^2 < 0$. This disproves simultaneous feasibility of the stated certificates, not existence of equilibria for every mechanism.

A payment failure and its repair. Write $\gamma > 0$ for Q's tax scale and $b_N = (N^2 - N + 1)/(N - 1)^2$, with $N \geq 2$. Direct collection of Q's equation eq40, at common prices and complementary capacity, gives

$$t_n^Q = \gamma b_N p^\top (x_n - c/N).$$

The printed coefficients also give $\sum_m a_m^n = \gamma c/[N(N - 1)]$, violating Q's P4(ii) for positive capacity (entries 5-7). Ada's counterexample takes $N = 3$, $\gamma = c = 1$, $X_n = [0, 1]$, and

$$V_1(x) = 10x - x^2/2, \quad V_2(x) = V_3(x) = 0.1x - x^2/2.$$

At $x = (1, 0, 0)$, $p = (9, 9, 9)$, own price derivatives vanish and own allocation cost derivatives are 0, 8.9, 8.9, with the latter at lower bounds. Each own cost Hessian is $\begin{pmatrix} 1 & -1/2 \\ -1/2 & 1 \end{pmatrix} \succ 0$, so these KKT conditions suffice for a Nash equilibrium. Nevertheless, $t_1^Q = 10.5 > V_1(1) = 9.5$: individual rationality fails. Emmy checked the same failure with coefficient 1 instead of 0.1 for the other valuations (entries 6, 7, 9).

Ada's opponent-only adjustment, confirmed by Emmy, is

$$t_n^{\text{corr}} = t_n^Q + r_n, \\ r_n = \gamma(b_N - 1)(\bar{p}_{-n}^{\text{av}})^\top \left(\sum_{j \neq n} x_j - \frac{N-1}{N} c \right), \quad \bar{p}_{-n}^{\text{av}} = \frac{\sum_{j \neq n} p_j}{N-1}.$$

It preserves all best responses. At common complementary prices, $t_n^{\text{corr}} = \lambda^\top (x_n - c/N)$, with $\lambda = \gamma p$, so equilibrium budget balance is exact. For normalized concave valuations, $0 \in X_n$, nonnegative capacity and multipliers, planner stationarity and the supporting-gradient inequality give

$$V_n(x_n) \geq \lambda^\top x_n, \quad V_n(x_n) - t_n^{\text{corr}} \geq \lambda^\top c/N \geq 0.$$

Thus the repair gives individual rationality when implementation holds; it does not restore Q's coefficient certificate or assert budget balance away from equilibrium (entries 8, 9, 17, 19).

Implementation without unique prices. Emmy's proof uses compact convex local sets, strongly convex costs $f_n = -V_n$, valid subgradient/normal-cone sum rules, and existence of a planner KKT pair. Let m be a common curvature lower bound, $\Pi = \mathbf{1}\mathbf{1}^\top/N$, $\Delta = I - \Pi$, and $\kappa = (2N - 1)/(N - 1)^2$; matrices act resourcewise. Differentiating eq40 and scaling price rows by $(N - 1)/N$ gives

$$G_x \in \partial f(x) + \gamma(\Pi - \kappa\Delta)p, \quad G_p = \gamma\Delta p - \gamma\Pi x + \frac{\gamma}{N}\mathbf{1}c.$$

For differences $u = x - x'$, $v = p - p'$, completion of squares yields

$$\begin{aligned} \langle G(s) - G(s'), s - s' \rangle &\geq m \|\Pi u\|^2 + (m - \gamma \kappa^2/4) \|\Delta u\|^2 \\ &\quad + \gamma \|\Delta v - (\kappa/2) \Delta u\|^2. \end{aligned}$$

Consequently the sufficient threshold is

$$m > \frac{\gamma(2N - 1)^2}{4(N - 1)^4}.$$

It also ensures own joint-cost convexity. A planner KKT pair gives a Nash equilibrium at $p_n = \lambda/\gamma$. Adding its variational inequality to that of any other equilibrium forces $u = 0$ and $\Delta v = 0$. The remaining stationarity conditions identify common prices as planner multipliers. Thus allocations are unique, while prices need not be (entries 4, 9, 14, 17, 19; Emmy's implementation proof).

The statistical surrogate and economic errors. Use upper-tail probability β_n , bounded losses $|J_n| \leq U_n$, samplewise convexity, Lipschitz constants L_n , and fresh iid samples from a query-independent distribution. Set

$$C_n = \text{CVaR}_{\beta_n}(J_n), \quad H_{n,s} = \mathbb{E} \hat{C}_{n,s}, \quad e_n = \frac{U_n}{\beta_n} \sqrt{\frac{2\pi}{s_n}}.$$

The threshold representation gives $\mathbb{E} \inf_{\tau} \hat{\phi} \leq \inf_{\tau} \mathbb{E} \hat{\phi}$, hence $H_{n,s} \leq C_n$. P's integrated Dvoretzky–Kiefer–Wolfowitz bound gives the other side, $C_n - e_n \leq H_{n,s}$. Centered ball smoothing and Jensen's inequality therefore imply

$$-e_n \leq \tilde{C}_n(x) - C_n(x) \leq L_n \delta_n, \quad w_n = e_n + L_n \delta_n.$$

These are bounds between deterministic expected functions, requiring no expected supremum of empirical errors (Ada, entry 13; Emmy's check, entry 17).

Normalize true and surrogate valuations at zero: $V_n(x) = C_n(0) - C_n(x)$ and analogously for $\tilde{V}_n$. For satisfaction data use loss $J_n = -\psi_n$; deterministic taxes enter additively by cash translation. At a corrected surrogate equilibrium, the oscillation w_n bounds actual unilateral regret and individual-rationality shortfall. Comparing surrogate optimality at $\tilde{x}$ and a true optimum x^* gives

$$\sum_n C_n(\tilde{x}_n) - \sum_n C_n(x_n^*) \leq \sum_n w_n, \quad \|\tilde{x} - x^*\|^2 \leq \frac{2 \sum_n w_n}{m_0},$$

when total population cost has curvature $m_0 > 0$. Budget balance stays exact. Applied to P with its additional domain-comparator term, the same value argument gives limiting error of order $O(\delta_{\max} + s_{\min}^{-1/2})$; it does not eliminate inverse-radius variance in finite-time learning (entries 13, 17 and Ada's note).

Local queries preserving zero. The late continuation resolves the local boundary objection. Given a known relative interior ball $a_n + \rho_n \mathcal{B} \subset X_n$, query at

$$q_n(x, u) = (1 - \varepsilon_n)x + \varepsilon_n a_n + \delta_n u, \quad 0 < \delta_n < \varepsilon_n \rho_n.$$

This is a convex combination of x and an interior-ball point, so every query is locally feasible. Decisions still include zero. For ball direction v , define $K_n(x) = E \hat{C}_{n,s}(q_n(x, v))$. P's sphere identity and the chain rule give the unbiased oracle

$$\hat{g}_n^K(x) = \frac{(1 - \varepsilon_n)d_n}{\delta_n} \hat{C}_{n,s}(q_n(x, u))u.$$

Here directions and dimension are relative to the local affine hull. Samplewise curvature m_n becomes $m_n(1 - \varepsilon_n)^2$, whose minimum must exceed the weighted threshold above. With $R_n = \sup_{x \in X_n} \|x - a_n\|$, the error interval is

$$-e_n - L_n \varepsilon_n R_n \leq K_n - C_n \leq L_n \delta_n + L_n \varepsilon_n R_n.$$

All economic bounds hold with $w_n^K = e_n + 2L_n \varepsilon_n R_n + L_n \delta_n$. Taking $\delta_n = \varepsilon_n \rho_n / 2$ gives order $O(s_n^{-1/2} + \varepsilon_n)$. This requires fresh samples at arbitrary locally feasible queries and guarantees no joint capacity feasibility of experiments (Ada, entries 20, 24; Emmy, entries 22, 26).

A changed learner with a proof. For fixed surrogate parameters, assume a compact product strategy set $\mathcal{S}$ with a price box containing a surrogate planner KKT equilibrium. The preceding comparison proves that all boxed equilibria retain the implemented allocation and complementary common prices; the reference price need not be strictly inside the box (entries 17, 24). With normal cone $N_{\mathcal{S}}$, use

$$R(z) = (I + \eta(\tilde{G} + N_{\mathcal{S}}))^{-1}z, \quad \eta > 0.$$

The inner operator $\tilde{G}(s) + \eta^{-1}(s - z)$ is strongly monotone. P's oracle, analytic payment gradients, projection, and aggregate messages give, for appropriate diminishing inner steps, $E[\|s_T - R(z)\|^2 \mid z] \leq C/(T+1)$. Taking $T_k = \lceil (k+1)^{2+\zeta} \rceil$, $\zeta > 0$, makes expected norm errors summable, hence almost surely summable. Firm nonexpansiveness of the resolvent gives convergent distances and vanishing residual; compactness identifies an equilibrium limit. Boundedness then gives mean-square convergence, with deterministic limiting allocation and possibly random common price (entries 11, 17, 19, 24; Emmy's full recurrence proof).

Objections and their disposition

The LMI contradiction and original individual-rationality failure remain objections to Q as printed; the replacement arguments do not validate those claims. Nor does weighted monotonicity prove Q's original proximal best-response (PBR) iteration nonexpansive. Emmy's interior quadratic example

with $N = 3$, $m = \gamma = 1$ has consensus-price column

$$\begin{pmatrix} -(\mu + 1)/D \\ 1 - 1/(2D) \end{pmatrix}, \quad D = (\mu + 1)^2 - 1/4,$$

whose squared norm is $1 + 1/(2D^2) > 1$ for every $\mu > 0$. Ada independently checked it. This refutes automatic Euclidean nonexpansiveness, not every possible convergence proof (entries 10, 14).

Expected curvature in Q does not ensure CVaR curvature; samplewise strong convexity is an added assumption (entries 9, 19). Full-message uniqueness also fails: with $N = 3$, $X_n = [0, 1]$, $c = 3/2$, $\gamma = 1$, equiprobable $Z = \pm 1$, and tail mass $1/2$,

$$J_n(x, Z) = x^2 - 3x + Z(x - 1/2), \quad C_n(x) = x^2 - 3x + |x - 1/2|.$$

The unique allocation is $x_n = 1/2$, but $\partial C_n(1/2) = [-3, -1]$, allowing every common price in $[1, 3]$ (entries 14, 17). The robust conclusion is allocation convergence. The boundary objection is resolved only locally under the known-ball oracle model, as the final checks explicitly stress (entries 24, 26).

Outside sources used

The peers' literature searches found Wang, Shen, and Zavlanos, *Risk-Averse No-Regret Learning in Online Convex Games*, ICML 2022 (PMLR 162, [wang22w](#)), and Cui and Shanbhag, [arXiv:2104.07860](#) (entry 12). These already cover empirical-CVaR one-point game learning and stochastic proximal methods in adjacent settings. Ada's note, referenced by entry 15, additionally records *Learning of Nash Equilibria in Risk-Averse Games*, [arXiv:2403.10399](#), and *Sample average approximation of conditional value-at-risk based variational inequalities*, DOI [10.1007/s11590-023-01996-9](#). The searches establish relevant prior work, not absence of a matching result. Ada explicitly reports checking landing pages and abstracts rather than full proofs. These sources constrain novelty claims; the algebra and changed-learner arguments above are supplied in the notes.

What remains open

The evidence for the algebraic corrections and conditional fixed-surrogate theorem is substantive, with independent exact-arithmetic checks recorded in entries 14, 17, 24, and 26. The material is thin on publication novelty and practical efficiency: no complete closest-literature comparison, efficient total oracle complexity, or stochastic performance benchmark is supplied. Convergence of original VS-PBR, an exact population-risk schedule with changing batches and radii, and jointly capacity-feasible exploratory queries remain unproved (entry 26). Fixed-surrogate convergence does not settle them.

The smallest next step

For the immediate publication question, read the closest full papers listed above and compare their assumptions and conclusions against the corrected payment formula, exact equilibrium transfer properties, and oscillation-based economic errors. This is the smallest missing check identified by both notes that could establish whether the supported package adds a distinct result (entries 15, 19, 26). It would settle that narrower novelty comparison; the separate algorithmic and joint-query questions still require their own proofs, and the record supplies no single step that settles all of them.